

Komal Ravindra Walhe

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Education

Rowan University

Master of Science in Data Science

Glassboro, N.J

Expected May 2026

- GPA: 3.89/4.0
- Relevant Coursework: Data Mining, Visual Analytics, Big Data, MLOps, Advanced DBMS, Patient Data Privacy and Ethics.

Modern Education Society's Wadia College of Engineering

Pune, India

Bachelor of Engineering in Computer Engineering

2019–2023

- GPA: 9.09/10.0 — Honors in AI & Machine Learning
- Relevant Coursework: Machine Learning, Data Structures, Data Science and Big Data Analytics, OS, Cloud Computing.

Skills

- **SQL:** Joins, CTEs, Window Functions, Aggregate Functions, PostgreSQL
- **Databases:** MongoDB
- **Data Visualization:** Tableau, Power BI, Streamlit, Plotly, Matplotlib, Seaborn.
- **Data Analysis:** Python (Pandas, NumPy, Polars), Excel (Pivot Tables, VLOOKUP, Power Query).
- **Technical Concepts:** Statistical Analysis, Data Cleaning/Wrangling, Data Validation, Predictive Modeling.
- **Tools & Cloud:** AWS (S3, EC2), Docker, Git, Jira, Pydantic, MLflow.

Technical Projects

Chicago Transit Analytics & Prediction — *Python, Polars, MongoDB, Streamlit*

- Built an end-to-end analytics pipeline using Medallion Architecture to process **800,000+ raw records**.
- Achieved a **20% reduction in data noise** by implementing Polars-based cleaning to generate 644,000 high-quality records.
- Developed an interactive Streamlit dashboard to visualize peak traffic times and popular community areas for urban planning.
- Optimized query performance by creating MongoDB indexes on high-traffic fields, reducing dashboard load times by **35%**.
- Performed exploratory data analysis (EDA) to identify traffic patterns and peak usage trends.

Stroke Risk Clinical Insights Dashboard — *Python, SHAP, Streamlit, Scikit-learn*

- Analyzed clinical datasets where only 4.87% of patients had strokes; used SMOTE to balance classes for **improved pattern recognition**.
- Used **SHAP (Explainable AI)** to identify top risk factors (age, glucose levels), translating complex models into clear visual insights.
- Developed a web-based interface for healthcare professionals to assess patient risk factors in real-time.

Insurance Risk Engine Validation — *Python, Pydantic, Pandas, Pytest*

- Built an automated tool to process **10,000+ records**, calculating risk scores based on actuarial rules and claims history.
- Implemented strict data validation using **Pydantic**, automatically rejecting invalid records to ensure high data integrity and reduce invalid entries.
- Integrated professional logging and unit testing with Pytest, reducing manual debugging time by **50%**.

Experience

Accenture (via Forage)

Remote

Data Analytics Virtual Experience

2024

- Analyzed large-scale datasets to identify top content categories for a social media client, improving content strategy insights.
- Created data-driven visualizations to communicate trends to stakeholders, aiding in improved client engagement strategy through actionable recommendations.

1Stop.ai

Remote

Machine Learning Intern

2022

- Performed exploratory data analysis (EDA) to uncover patterns in text and image data for classification tasks.
- Improved model training efficiency by **15%** through optimized data pre-processing and feature extraction techniques.

Leadership & Activities

- **NJ CostSync AI Hackathon:** Pitch Lead; translated technical data architectures into business-ready presentations for judges.
- **IBM SkillsBuild AI Lab:** Documenting "As-Is" vs "To-Be" data workflows for AI agent deployment.